

H0 — decision summary

llama33-70b (ctx 131,072, 5,120 heads)

HEADS IN BAND 28.4% (gain over best corner $\geq 2.0\times$ at 3 b/token)

ROUTED GAIN 1.53x (geometric mean of $\max(\text{gain}, 1)$ over all heads)

in-band heads median 3.2x p90 6.2x

ladder width median 1.96 b IQR [1.11, 2.79] 64% of heads above 1.4

tau median 1.36 excl. sinks 1.36

top-1 weight median 0.0014 95% mass in 78498 tokens (65.1% of ctx)

B=2b vs uniform 1.4x vs eviction 4.9x vs BEST corner 1.3x evicts 39%

B=3b vs uniform 1.1x vs eviction 7.2x vs BEST corner 1.0x evicts 21%

quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.488 [H2 answered]

heads where eviction beats a 1-bit tier: 28%

heads where eviction beats a 2-bit tier: 5%

linearization check (predicted/measured gain @3b): 4.89 (1.0 = exact)

VERDICT: NARROW – 28% of heads are in band (1.53x routed). Real but partial. Ship the router as part of the method and make the regime bound

qwen3-30b-a3b-2507 (ctx 131,072, 1,536 heads)

HEADS IN BAND 14.8% (gain over best corner $\geq 2.0\times$ at 3 b/token)

ROUTED GAIN 1.29x (geometric mean of $\max(\text{gain}, 1)$ over all heads)

in-band heads median 2.3x p90 3.0x

ladder width median 4.41 b IQR [3.78, 5.50] 100% of heads above 1.4

tau median 3.05 excl. sinks 3.04

top-1 weight median 0.2015 95% mass in 3192 tokens (2.6% of ctx)

B=2b vs uniform 54.8x vs eviction 1.3x vs BEST corner 1.3x evicts 68%

B=3b vs uniform 31.2x vs eviction 1.1x vs BEST corner 1.1x evicts 54%

quantizer alpha @2b 0.940 (1.0 = no temperature distortion); spearman@2b 0.468 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

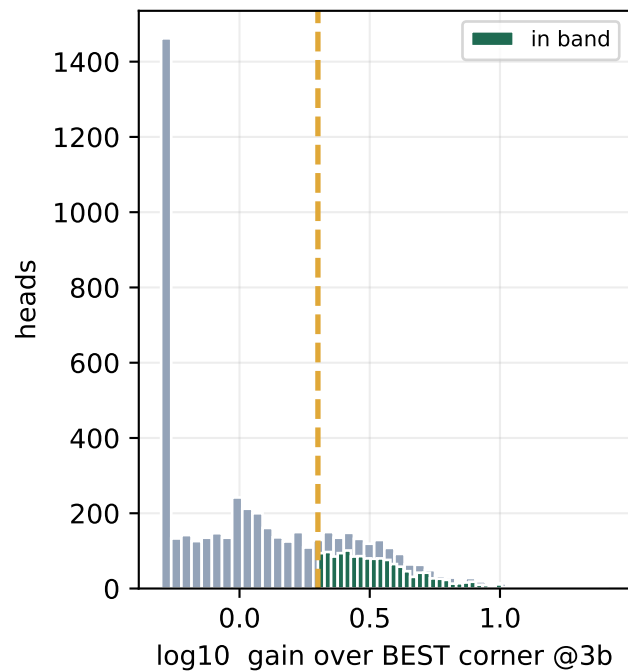
heads where eviction beats a 2-bit tier: 77%

linearization check (predicted/measured gain @3b): 0.93 (1.0 = exact)

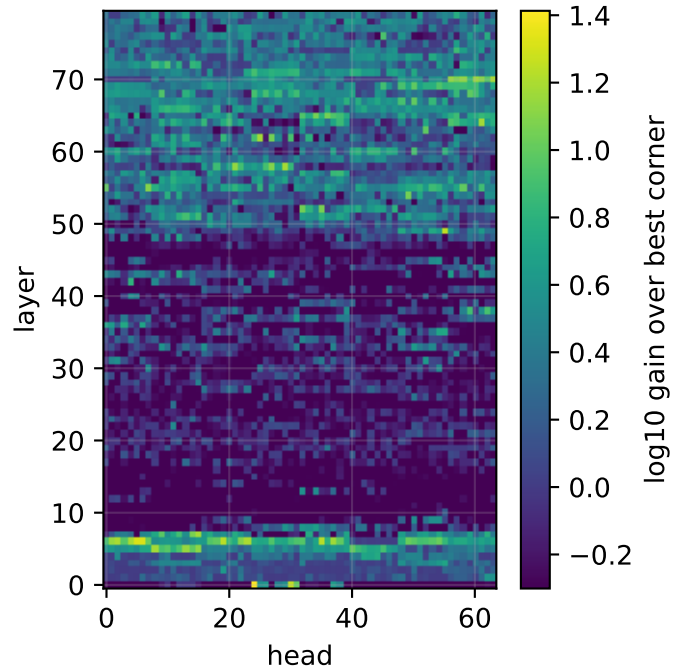
VERDICT: STOP – Only 15% of heads sit in the productive band and routing buys 1.29x overall. Uniform quantization or eviction already captured

llama33-70b — ctx 131,072

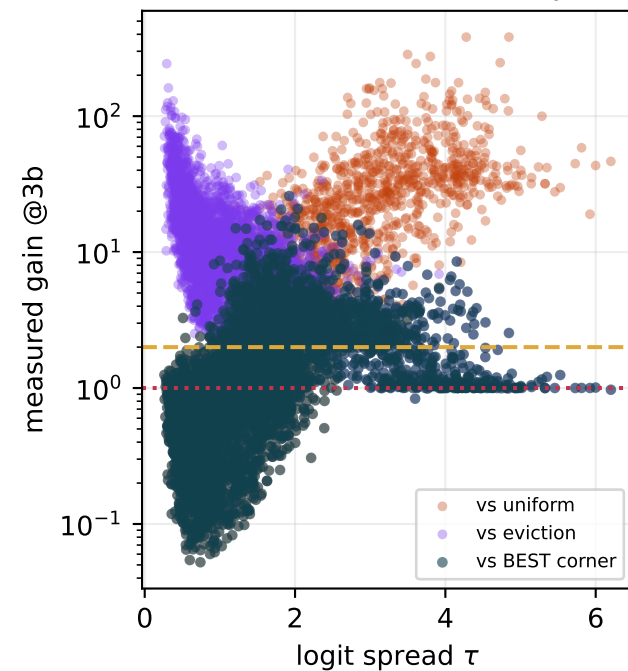
HEADS IN BAND: 28%



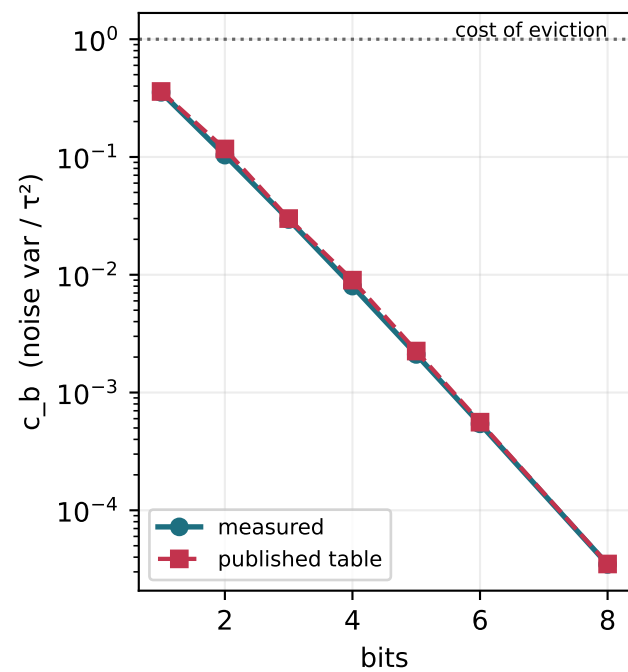
Which heads the router sends to SIEVE



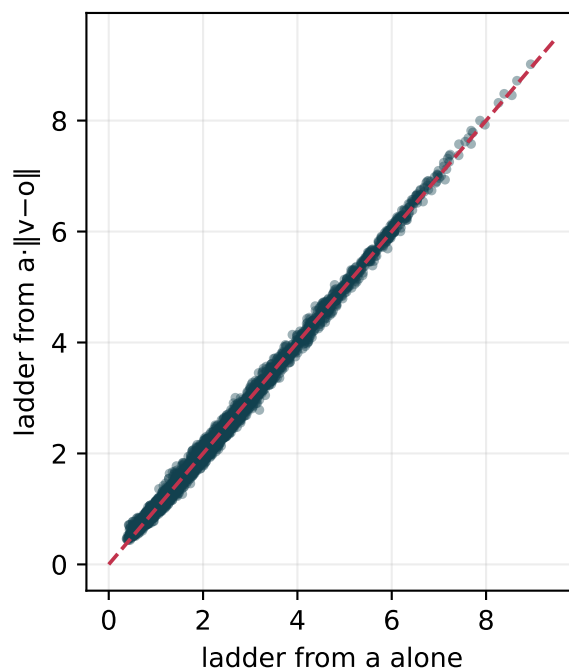
Where does the interior actually win?



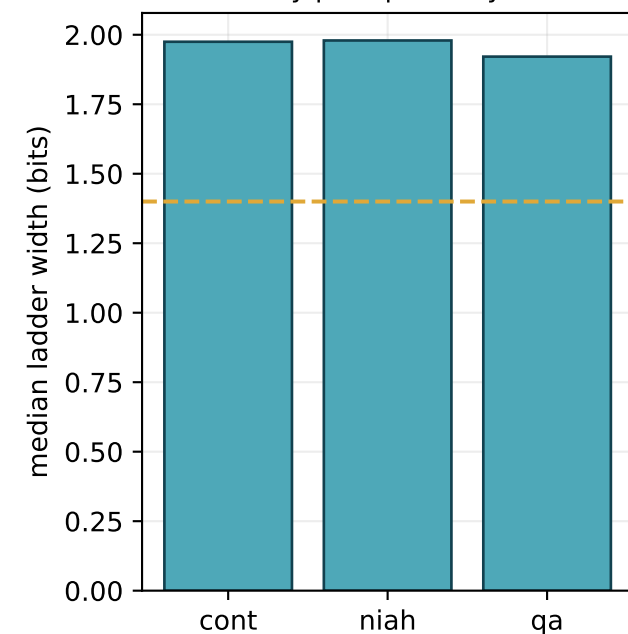
Measured vs assumed noise



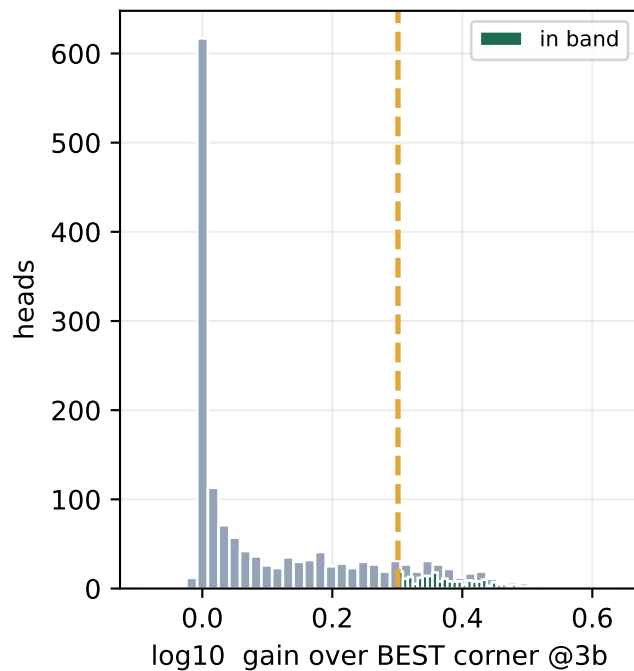
Does the value term matter?



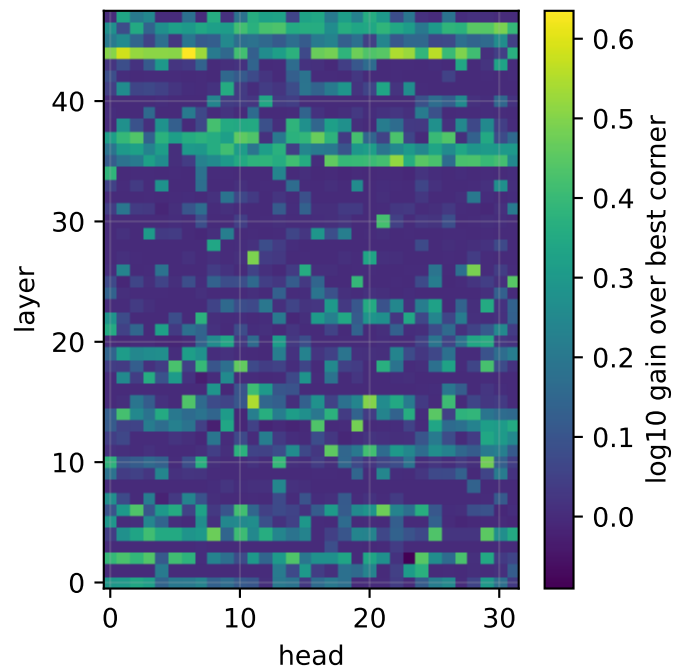
By prompt family



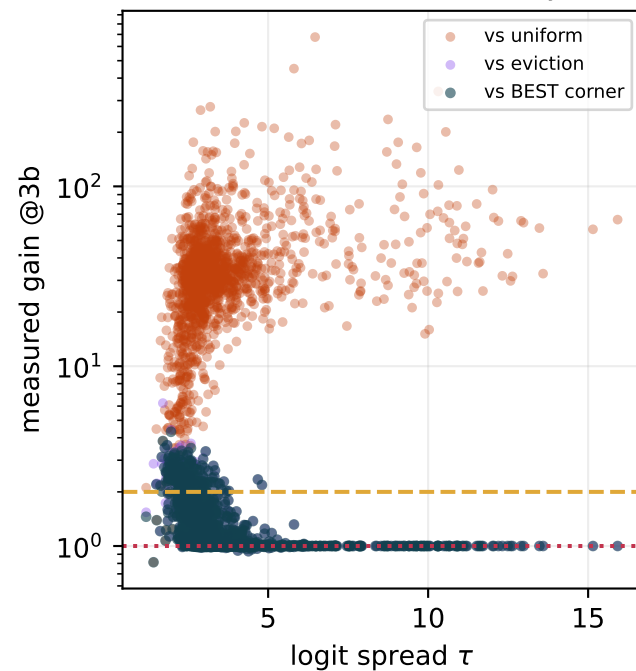
HEADS IN BAND: 15%



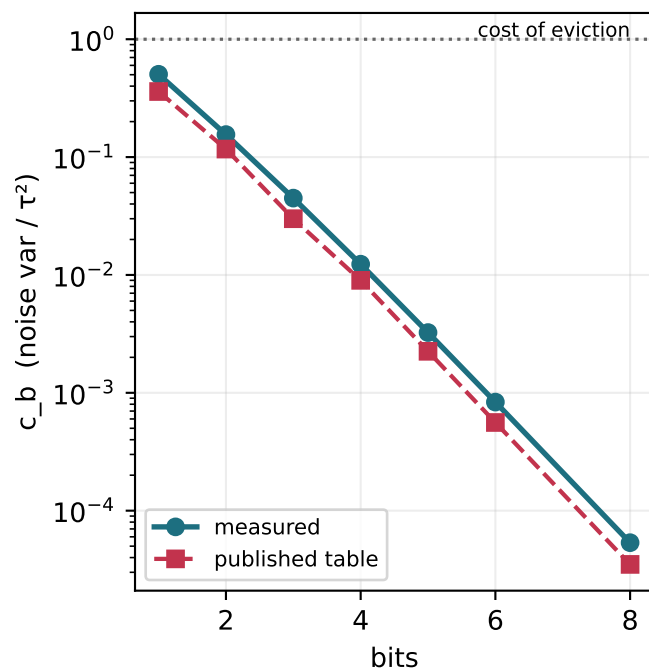
Which heads the router sends to SIEVE



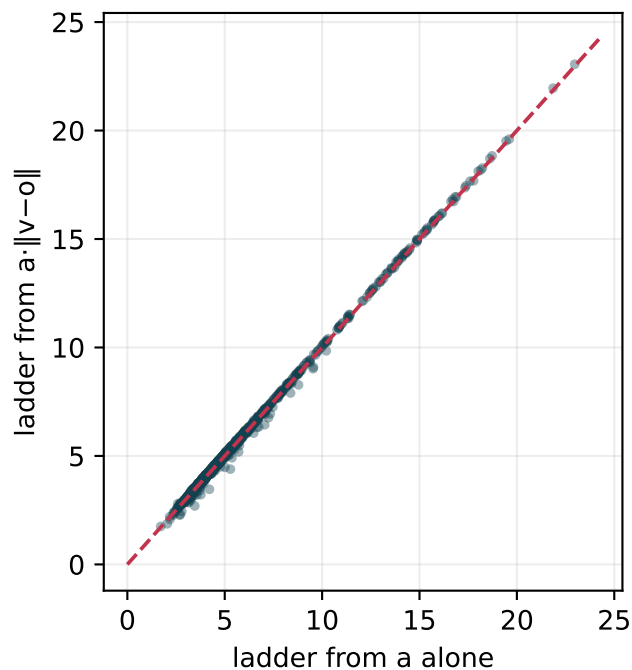
Where does the interior actually win?



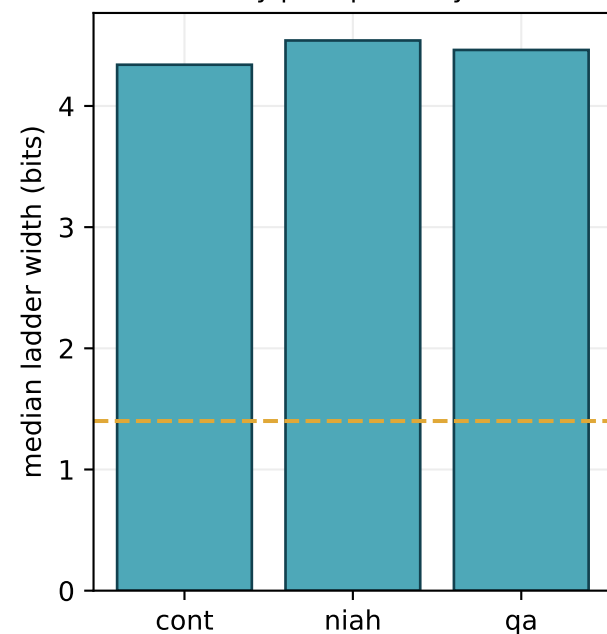
Measured vs assumed noise



Does the value term matter?



By prompt family



Cross-model comparison

